

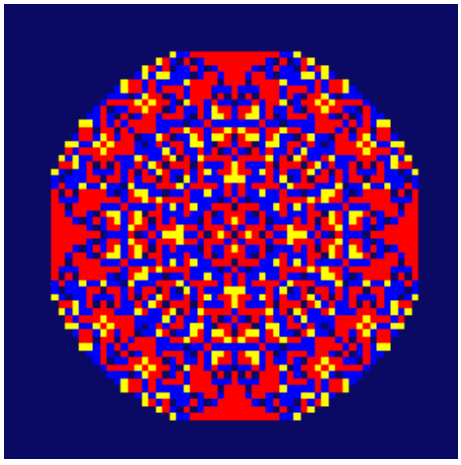
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Final Report

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Chapter 1

Introduction

Complex systems are common in both nature and society and are closely related to our daily lives. Many significant large-scale phenomena, such as climate change, forest fires and pandemics, can be understood and studied as complex systems. In general, a complex system can be described as a large, coupled, nonlinear spatial system in which components interact with each other. Because of these interactions, the macroscopic behaviour of the system can be difficult to recognise from the microscopic model [5].

This report will focus on self-organised criticality (SOC) and the associated non-Gaussian statistics in complex systems. In particular, I will use the Bak–Tang–Wiesenfeld (BTW) sandpile model, a representative model exhibiting self-organised criticality.



Figure 1.1: A melting iceberg in Arctic waters [1].



Figure 1.2: A wildfire in California [8].

1.1 Self-Organised Criticality (SOC) and Scale Invariance

According to the research of Bak, Tang, and Wiesenfeld, dynamical systems with extended spatial degrees of freedom naturally evolve into a self-organised critical state, which is barely stable. In such a state, small perturbations at a certain location may render the neighbouring sites unstable, and the perturbations may spread to the neighbours in a chain reaction [2]. Self-organised means that the system can naturally evolve to the critical state without detailed specification of the initial conditions. A critical state is a state with no characteristic time or length scales, exhibiting scale invariance [3].

The cover image was generated using WebGL Sandpiles: `https://people.reed.edu/~davidp/web_sandpiles/`.

Scale invariance is usually defined as the structure or property of a system remaining unchanged as the scale changes. Mathematically speaking, it can be described as

$$f(\lambda x) = C(\lambda) f(x)$$

for $\lambda > 0$, where $C(\lambda)$ is the scaling factor that does not depend on x [7]. This relation means that multiplying the input by a factor of λ preserves the functional form, while multiplying the output by a corresponding factor of $C(\lambda)$. This requirement constrains the functional form to a power law, usually written as $f(x) = Ax^{\alpha}$, where A and α are constants. This explains why phenomena in systems exhibiting scale invariance can be described by power laws [9, 11].

1.2 Bak–Tang–Wiesenfeld (BTW) model

The Bak–Tang–Wiesenfeld (BTW) sandpile model was introduced by Per Bak, Chao Tang and Kurt Wiesenfeld to explain self-organised criticality [2]. The BTW sandpile model is selected in this project because its simple local rules can lead to complex collective behaviour through interactions between sites.

In particular, the model consists of a two-dimensional grid of a chosen size. At each step, a grain of sand is added to a randomly chosen grid point. If the number of grains at a grid point reaches or exceeds four, the site topples, transferring one grain to each of its four nearest neighbours. Any grains that fall outside the grid are removed from the system. A new grain is added only after all toppling activity has ceased.

The model has extended spatial degrees of freedom, as each site of the two-dimensional grid has its own state, represented by the number of sand grains at that site. According to the toppling rule, a topple at one site affects its neighbours and may trigger further toppling, leading to a chain reaction. Additionally, sand addition, toppling, and open boundaries allow the system to self-regulate the accumulation and loss of sand grains, eventually reaching a state where average input balances average loss. This process illustrates how the system regulates itself without external fine-tuning. In the resulting state, the distribution of avalanche area exhibits approximate power-law behaviour, indicating scale invariance and criticality [2].

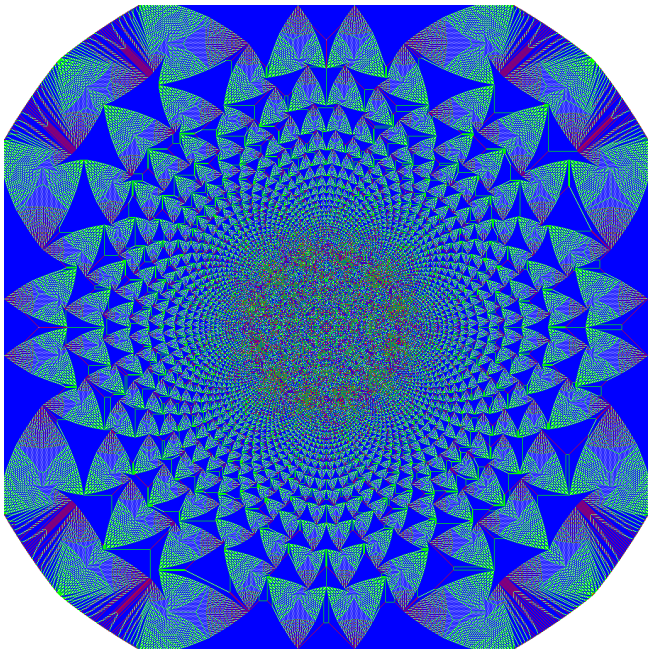


Figure 1.3: A BTW sandpile pattern in which the sand was dropped continuously onto the centre [10].

1.3 Aim

This project aims to study SOC and the associated non-Gaussian statistics in the basic BTW sandpile model, and to compare this model with a gradient-dependent extension. In particular, this project will mainly focus on investigating the steady-state behaviour of the basic model, analysing the distributions and correlations of four avalanche properties and examining the mathematical properties of the toppling rules. The behaviour and avalanche statistics of the basic model will then be compared with the gradient-dependent extended model.

Chapter 2

Methodology

2.1 Basic BTW Model Rules

As mentioned in the Introduction, the BTW sandpile model is defined on a two-dimensional grid of a preset size with sand grains added at randomly chosen sites, prescribed toppling rules and open boundaries.

The first step of simulating a basic BTW sandpile model is to initialise an empty two-dimensional grid of preset size $N \times M$. At each step, one grain of sand is added to a randomly chosen grid point. After sand addition, if a grid point contains four or more grains of sand, four grains are removed from it, and one grain is added to each of its neighbours: the grid points above, below, left and right. If a neighbour of the toppling site lies outside the grid, the grain transferred in that direction is removed from the system. Toppling continues until every grid point contains fewer than four grains of sand. The state and avalanche properties are then recorded. Finally, the sand-addition and toppling process is repeated until the preset number of grain additions is reached.

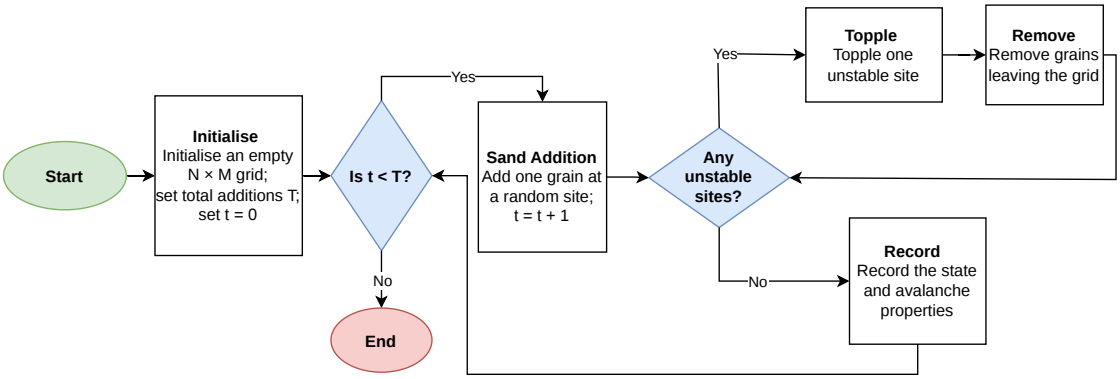


Figure 2.1: Flowchart of the basic BTW sandpile simulation.

Mathematically, we can consider the grid as an $N \times M$ matrix. Initially, all elements in the matrix are zero. When a grain of sand is added, the corresponding element in the matrix increases by one.

Define $p_t(i, j)$ to denote the number of sand grains in row i, column j, at time t. If $p_t(i, j) \geq 4$, the following updates are applied:

$$\begin{aligned} p_t(i, j) &= p_t(i, j) - 4, \\ p_t(i \pm 1, j) &= p_t(i \pm 1, j) + 1, \\ p_t(i, j \pm 1) &= p_t(i, j \pm 1) + 1. \end{aligned}$$

For example:

$$\begin{pmatrix} 0 & 3 & 1 \\ 1 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 3 & 1 \\ 1 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 4 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

First, one grain of sand is added to position $(2, 2)$. This grid point now contains four grains of sand, so toppling occurs. Four grains of sand are removed from position $(2, 2)$, and the number of grains at each of its neighbours, positions $(1, 2)$, $(2, 1)$, $(3, 2)$ and $(2, 3)$, increases by one. Now, position $(1, 2)$ contains four grains of sand, so toppling happens again. Since one of the neighbours of position $(1, 2)$ is outside the matrix, one grain of sand is lost from the system.

It is important for the table to have finite open boundaries. These allow sand to leave the system, preventing the total number of sand grains from increasing indefinitely. If we have an infinite lattice but drop sand only on a finite subset of the lattice, sites outside may also topple and transfer sand back into it. Therefore, the subset cannot be treated as a finite system with open boundaries.

2.2 Measured Properties

In this project, I will record and analyse the following avalanche properties:

Topples The total number of toppling events during an avalanche.

Area The number of distinct sites that topple during an avalanche.

Loss The number of sand grains lost through the open boundaries during an avalanche.

Length The maximum Manhattan distance from the initial toppling site to any site toppled in the avalanche.

The avalanche length measures how far the toppling activity spreads from the initial toppling site. Manhattan distance, calculated as $|x_1 - x_2| + |y_1 - y_2|$ between sites (x_1, y_1) and (x_2, y_2), is used because sand is transferred only to the neighbours above, below, left and right.

Another important property that will be focused on is the average height of the pile. For an $N \times M$ grid, its average height is calculated as:

$$\langle p_t \rangle := \frac{1}{NM} \sum_{i,j} p_t(i,j)$$

2.3 Simulation Setup

To simulate the BTW model using Python, I initialise each pile as a 50×50 matrix with all elements set to zero. A total of 100,000 grains of sand are added to each pile, one at a time. I independently simulate five piles to assess the variability between runs. To ensure the reproducibility of the random selection, I set the seed to 1643210.

I first define a sand addition function that takes a matrix as input, then randomly selects one element from it and increments its value by 1.

The toppling rule is implemented by taking a matrix as input and using a while loop that continues until no element in the matrix has a value greater than or equal to 4. Inside the while loop, when a position with a value greater than or equal to 4 is found, its value is decreased by 4. For each neighbour, an if statement checks whether it lies within the matrix. If it is, its value is increased by one. Otherwise, the loss count is increased by one. The function records the row and column indices of all toppling events, the total loss, and the final matrix after all toppling activity has ceased. Additionally, I define three functions to calculate topples, area, and length using the output from the toppling function. If no toppling occurs after a grain addition, all four avalanche properties are recorded as zero.

For each pile, the sand-addition and toppling functions are called repeatedly 100,000 times. After each grain addition and toppling, the four avalanche properties and the average height are recorded. Then I generate a plot of the average pile height against the number of grains added. Since all piles are initially empty, they require a number of grain additions to reach a steady state. The plot is used to identify when the average pile height begins to fluctuate around a stable level. Based on this observation, the first 10,000 records from each pile are removed, and the remaining records are used for further analysis of the avalanche properties. Using these remaining records, I generate linear-scale, semi-log, and log-log plots for Topples, Area, Loss, and Length to explore their distribution shapes and identify candidate distributions for fitting. I also plot heat maps of the final configurations to compare the spatial distributions of sand heights in the basic and extended models.

2.3.1 Statistical Analysis

Topples

For the Topples property, I attempt to fit a power law. In particular, I first exclude zero values since $\log(0)$ is undefined. Let x denote a positive number of topples and y be its corresponding observed frequency. The power-law relationship can be written as

$$y = Ax^{\alpha},$$

and taking logarithms gives

$$\log(y) = \log(A) + \alpha \log(x).$$

Therefore, I use `np.polyfit` to perform a fit of $\log(y)$ against $\log(x)$ and estimate A and α. Then I plot the fitted curve and the observed frequencies on the same log-log plot. This estimation is performed separately for each pile.

Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are used to evaluate the differences between the observed and fitted frequencies:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|,$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

where n is the number of distinct positive Topples values observed, y_i and $\hat{y}_i$ are the observed and fitted frequencies at each value, respectively. Both errors are calculated on the original frequency scale.

Area

Similarly, for the Area property, I first fit a power law, but the fitted curve does not describe the observed frequency distribution well. Therefore, I further check if the distribution can be expressed as a log-normal distribution.

The log-normal distribution can be expressed as:

$$\log(X) \sim N(\mu, \sigma^2)$$

where X represents Area. I first estimate μ and σ^2 from the logarithms of the positive Area values by

$$N = \sum_i y_i,$$

$$\hat{\mu} = \frac{\sum_i (y_i * \log(x_i))}{N},$$

$$\hat{\sigma}^2 = \frac{\sum_i \{y_i * [\log(x_i) - \hat{\mu}]^2\}}{N}.$$

Since the log-normal distribution is continuous, I approximate the frequency at each integer area value using the probability within a centred unit-width interval. The fitted frequency is calculated as

$$\hat{y} = N[P(X \le x + 0.5) - P(X \le x - 0.5)].$$

Similarly, the fitted frequencies are compared with the observed Area frequencies on the same log-log plot and are used to calculate MAE and RMSE.

Loss

For the Loss property, I first fit a geometric distribution that includes zero, with probability mass function:

$$\Pr(X = x) = p(1-p)^x, \qquad x = 0, 1, 2, \ldots,$$

where X represents Loss. Since the mean is $\mu = \frac{1-p}{p}$, I estimate p using the sample mean $\bar{x}$:

$$\hat{p} = \frac{1}{\bar{x} + 1}.$$

The fitted frequencies are then calculated as:

$$N = \sum_i y_i,$$

$$\hat{y} = N\hat{p}(1-\hat{p})^x.$$

Since the fitted curve does not describe the observed frequencies well on the semi-log plot, I then fit a geometric distribution to the positive Loss values, excluding zero.

For a geometric distribution that excludes zero, the probability mass function becomes:

$$\Pr(X = x) = p(1-p)^{x-1}$$

The mean is now $\mu = \frac{1}{p}$, so $\hat{p} = \frac{1}{\bar{x}}$, where $\bar{x}$ is calculated from the positive Loss values. I calculate the fitted frequencies using the updated probability mass function and the number of positive observations, and plot the fitted curve and the observed frequencies on the same semi-log plot. MAE and RMSE are also calculated and recorded.

I also try to fit a log-normal distribution, plot the fitted curve and calculate MAE and RMSE to compare the two distributions.

Length

For the Length property, I also fit a log-normal distribution to the observed frequencies. I use the same fitting and error calculation procedures as for the log-normal fit to Area and Loss.

Correlation

In order to explore the correlations between the four properties, I first create pairwise scatter plots using only data from the first pile, where the x-axis represents one property and the y-axis represents another. For each pile, I calculate the Spearman correlations between the properties and present them in a correlation matrix.

I choose Spearman correlation because it measures monotonic relationships between properties without requiring these relationships to be linear. It is calculated using the ranks of the observations.

Chapter 3

Abelian Property of the BTW Model

The toppling process is Abelian, which means that the order in which unstable sites topple does not affect the final stable state.

For example:

$$\begin{pmatrix} 0 & 4 & 1 \\ 1 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 5 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} 0 & 4 & 1 \\ 1 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 \\ 1 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

When two sites contain four or more grains, the order in which they topple first does not affect the final stable matrix.

3.1 Toppling Matrix

To examine why the toppling order does not affect the final stable state, I will first introduce a mathematical representation of the toppling operation, the toppling matrix $\Delta(x, y)$. This matrix is used to describe how the number of grains at each site y changes when an unstable site x topples.

The toppling matrix must satisfy the following four conditions:

$$\forall x, y \in L,\ x \neq y, \quad \Delta(x, y) = \Delta(y, x) \geq 0 \tag{1}$$

$$\forall x \in L, \quad \Delta(x, x) < 0 \tag{2}$$

$$\forall x \in L, \quad \sum_{y \in L} \Delta(x, y) \leq 0 \tag{3}$$

$$\sum_{x,y \in L} \Delta(x, y) < 0 \tag{4}$$

where L represents the set of all positions in the grid.

Condition (1) means that for two distinct sites $x, y \in L$, the number of sand grains transferred from x to y when x topples is equal to the number transferred from y to x when y topples. Additionally, these quantities must be non-negative. This equality is required to match the basic BTW rule. The non-negativity condition ensures that toppling at one position does not decrease the number of grains at its neighbour, so an already unstable position remains unstable until it topples.

Condition (2) ensures that the number of grains at the toppling site decreases when it topples. Otherwise, the site would remain unstable after each toppling, allowing it to topple indefinitely.

Condition (3) means that the total change in the number of sand grains caused by a toppling, including at the toppling site itself, must be non-positive. If this condition does not hold, a toppling increases the number of sand grains, and toppling may continue indefinitely rather than reaching a stable configuration.

Condition (4) ensures that there is at least one position where toppling causes sand to leave the system. Otherwise, no sand can leave the system through toppling. Once a stable pile contains $3NM$ grains, adding one more grain of sand at any position will cause toppling activity that cannot terminate.

For the BTW model, the toppling matrix is defined as follows:

$$\Delta(x,y) = \begin{cases} -2d, & x = y, \\ 1, & x \text{ and } y \text{ are nearest neighbours}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

where d represents the dimension of the grid. For example, for the two-dimensional grid used in this simulation, $-2d = -4$. The value $-2d$ represents the change in the number of grains at the toppling site, because a d-dimensional grid has $2d$ nearest-neighbour directions, with one grain transferred in each direction. When toppling occurs, the toppling site transfers one grain of sand to each nearest neighbour and does not transfer any grains to other sites during this single toppling. The stability threshold k is equal to $2d$.

All four conditions above are satisfied by this toppling matrix. Condition (1) is satisfied because, for every $y \neq x$, $\Delta(x,y)$ is either 1 or 0, both of which are non-negative. Additionally, if x is a nearest neighbour of y, then y is also a nearest neighbour of x, so $\Delta(x,y) = \Delta(y,x)$.

Since d represents the dimension of the grid and is a positive integer, $-2d$ is negative, satisfying Condition (2).

Condition (3) is also satisfied because each site has at most $2d$ nearest neighbours within L. Therefore,

$$\sum_{y \in L} \Delta(x,y) \leq -2d + 2d = 0.$$

Since sites on the open boundaries have fewer than $2d$ nearest neighbours within L, their row sums are negative, while those of interior sites are zero, making the total sum strictly negative. Therefore, Condition (4) is also satisfied.

3.2 Toppling Operators

According to the definition of the toppling matrix, the toppling operator T_x, which maps a configuration p to a new configuration p', is defined as follows:

$$(T_x p)(y) := \begin{cases} p(y) + \Delta(x,y), & p(x) \geq -\Delta(x,x), \\ p(y), & \text{otherwise.} \end{cases} \tag{6}$$

Consider two distinct unstable sites x and x', with $p(x) \geq k$ and $p(x') \geq k$. By Condition (1), toppling at either site does not decrease the number of grains at the other site. For any $y \in L$,

$$(T_x T_{x'} p)(y) = p(y) + \Delta(x', y) + \Delta(x, y) = (T_{x'} T_x p)(y).$$

Therefore, T_x and $T_{x'}$ commute for unstable configurations.

3.3 Stabilisation and Uniqueness

The stabilisation operator is introduced to map an unstable configuration to a stable configuration and is denoted by $\mathbb{T} : U_L \to S_L$, where U_L represents the space of all possible height configurations and $S_L \subset U_L$ is the subset of stable height configurations. The stabilisation operator can be represented as:

$$\mathbb{T} = \prod_{i=1}^{N} T_{x_i}$$

where N is the total number of topplings required to stabilise the configuration, and x_i denotes the site of the ith toppling.

Consider an unstable configuration p that is stabilised after N topplings:

$$\mathbb{T}p = \left(\prod_{i=1}^{N} T_{x_i} \right) p.$$

Different legal toppling sequences starting from p have the same initially unstable sites. Since an unstable site remains unstable until it topples, its toppling can be delayed and exchanged with other legal toppling operations to make the first operation the same in each sequence. This produces the same configuration, including any newly unstable sites. This process is repeated on the remaining operations until stability is reached. Therefore, all sequences produce the same final stable configuration, so $\mathbb{T}$ is well-defined.

Chapter 4

Results

As mentioned in the Methodology, the simulations are performed on five independent 50×50 matrices. For each matrix, a total of 100,000 grains of sand are added one at a time to randomly selected sites. Average height, Topples, Area, Loss, and Length are recorded after each addition and the completion of any resulting toppling activity.

4.1 Steady state and burn-in

Since the simulation starts from an empty matrix, the system requires a number of grain additions to reach a steady state. Therefore, I generate a plot of the average pile height against the number of grains added to estimate when the system reaches a steady state and exclude the records from the burn-in period.

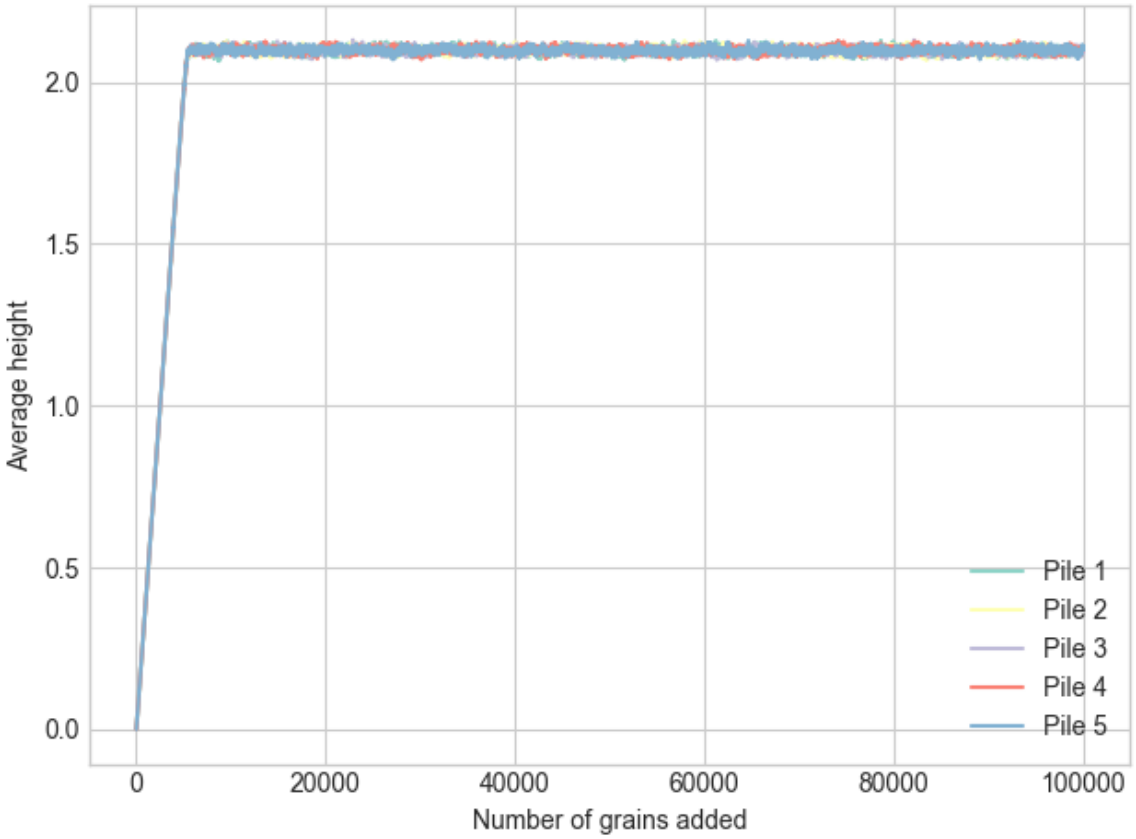


Figure 4.1: Average pile height against the number of grains added for five independent simulations.

As shown in Figure 4.1, the five piles show similar patterns. By about 10,000 grain additions, the average height has stabilised and fluctuates around 2.1. Therefore, I exclude the records from the first 10,000 additions as burn-in before further analysis.

4.2 Avalanche Property Distributions

After excluding the records from the first 10,000 additions, I use the remaining records to plot the observed frequencies of Topples, Area, Loss, and Length. I generate linear-scale, semi-log, and log-log plots for each of the four properties to explore their distribution shapes and identify candidate distributions for fitting. The linear-scale plots are shown in Figure A.1 in Appendix A.

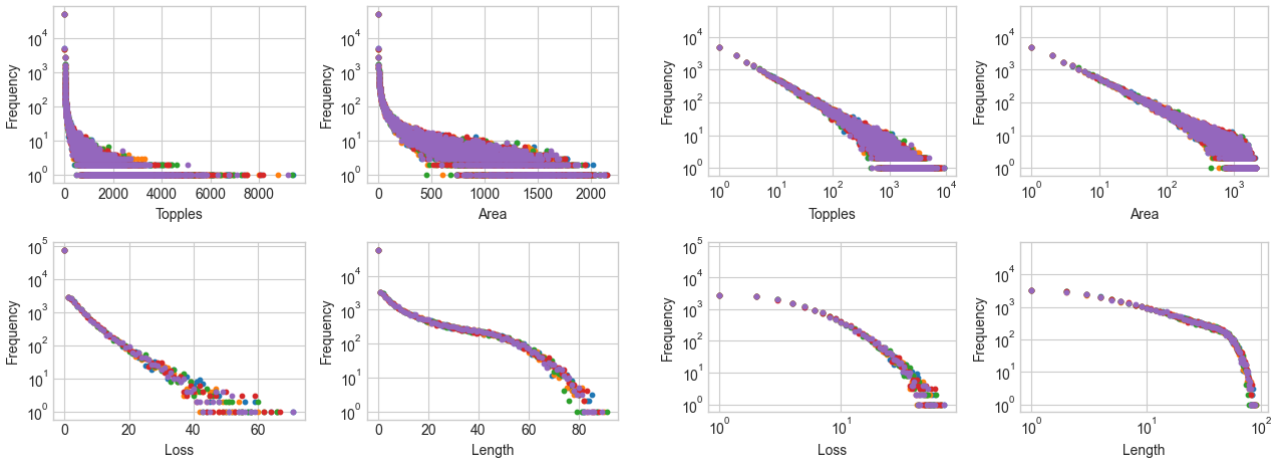


Figure 4.2: Observed frequencies of Topples, Area, Loss, and Length for five independent simulations on semi-log axes (left) and log-log axes (right).

As shown in Figure 4.2, the positive Loss values follow an approximately linear pattern on the semi-log plot, apart from deviations in the tail. Topples and Area show approximately linear patterns over part of their ranges on the log-log plot. However, Length does not follow a straight-line pattern on either plot.

Then I try to find a fit for the distribution for each of the avalanche properties.

4.2.1 Topples

Since Topples shows an approximately linear pattern on the log-log frequency plot, I consider a power law as a candidate distribution. I fit a power law to the observed frequencies and plot the fitted curve on the same scatter plot.

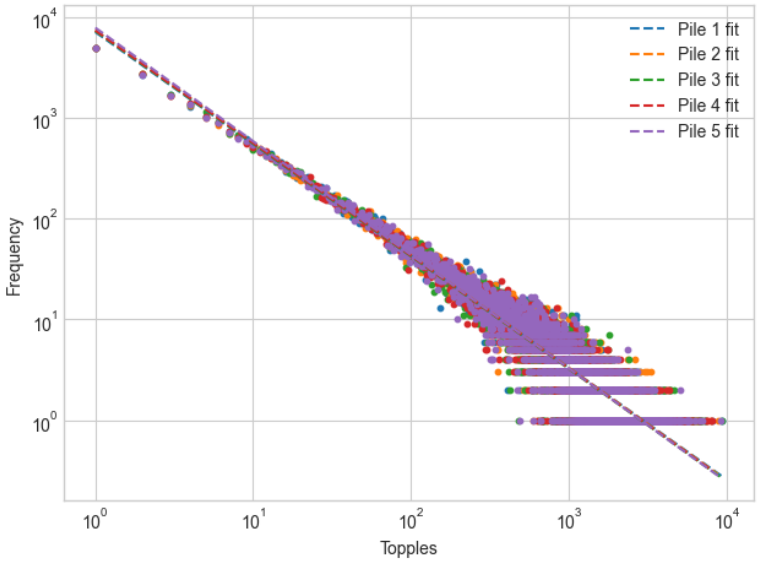


Figure 4.3: Observed positive Topples frequencies and fitted power-law curves for five independent simulations on log-log axes.

The five independent piles show similar patterns. The observed frequencies are distributed close to the fitted curves except in the tails. The estimated power-law exponent α, MAE, and RMSE for each pile are shown in Table 4.1.

Pile	α	MAE	RMSE
1	-1.1110	3.4764	47.1356
2	-1.1174	3.8661	55.3308
3	-1.1181	3.5590	52.4131
4	-1.1185	3.7551	55.4676
5	-1.1274	4.2641	64.2137

Table 4.1: Estimated power-law exponents and fitting errors for Topples.

The estimated exponents range from -1.1274 to -1.1110, showing little variation across the five piles. Additionally, MAE ranges from 3.4764 to 4.2641, while RMSE ranges from 47.1356 to 64.2137. The substantially larger RMSE indicates that some observed frequencies deviate considerably from the fitted values.

Overall, the observed Topples frequencies suggest approximate power-law behaviour, with fitted exponents close to -1.12, although deviations are present in the tails.

4.2.2 Area

Since the observed Area frequencies show an approximately linear pattern on the log-log plot, I first fit a power law, as for Topples. Similarly, I plot the fitted curve on the scatter plot and calculate MAE and RMSE. The fitted curves and fitting results are shown in Figure A.3 and Table A.1. The fitted curve does not describe the trend of the observed frequencies well. I then fit a log-normal model to the observed frequencies and plot the fitted curve on the scatter plot.

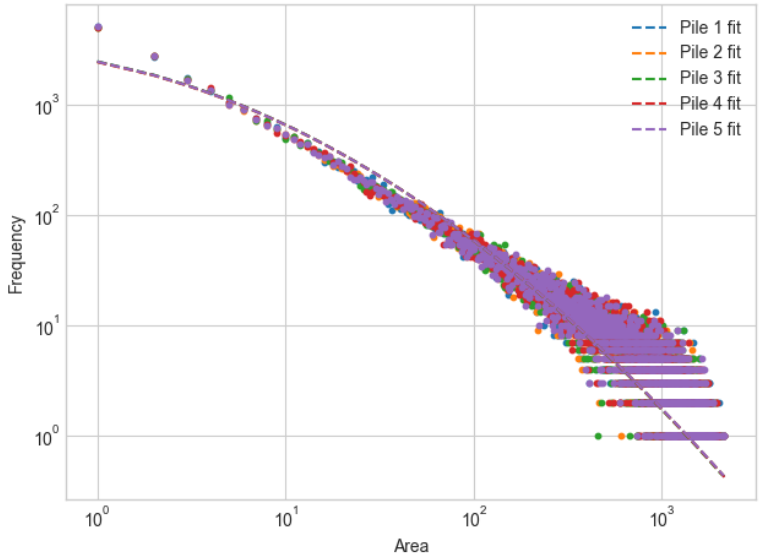


Figure 4.4: Observed positive Area frequencies and fitted log-normal curves for five independent simulations on log-log axes.

The fitted log-normal curves appear to describe the observed Area frequencies better. The estimated mean μ and variance σ^2 of log(Area), with MAE and RMSE for each pile, are shown in Table 4.2.

Pile	μ	σ^2	MAE	RMSE
1	3.2215	4.8049	7.1053	68.8535
2	3.2310	4.8103	7.2480	69.0048
3	3.2138	4.7811	7.0596	66.6385
4	3.2267	4.7858	7.0092	66.9160
5	3.2295	4.8169	7.2742	69.8435

Table 4.2: Estimated log-normal parameters and fitting errors for positive Area values.

It is worth noting that the MAE and RMSE for the power-law fit are about 16 and 257, respectively, which are much larger than those for the log-normal fit. Therefore, the log-normal model provides a better fit to the observed Area frequencies, with estimated parameters $\mu \approx 3.22$ and $\sigma^2 \approx 4.8$.

4.2.3 Loss

According to the semi-log plot of Loss frequencies above, the positive Loss values follow an approximately linear pattern. Therefore, I first fit a geometric distribution including zero. However, the fitted curves differ considerably from the observed frequencies, as shown in Figure A.4. I then fit a geometric distribution to only the positive Loss values, which gives a better fit.

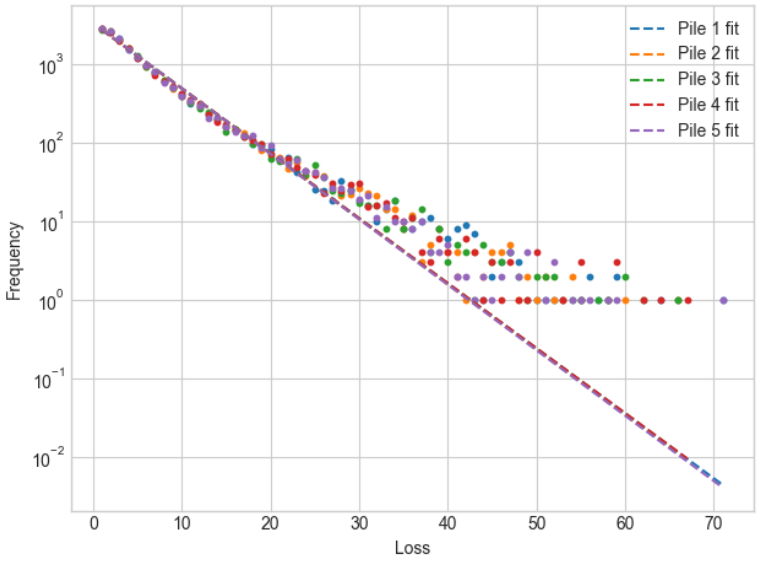


Figure 4.5: Observed positive Loss frequencies and fitted geometric curves for five independent simulations on semi-log axes.

Similarly, I calculate and record the estimated parameter p, MAE, and RMSE for each pile, as shown in Table 4.3.

Pile	p	MAE	RMSE
1	0.1733	27.0177	59.4437
2	0.1741	28.8757	67.8951
3	0.1735	27.2988	62.3667
4	0.1732	28.6606	60.1797
5	0.1744	30.1436	68.4744

Table 4.3: Estimated geometric distribution parameters and fitting errors for positive Loss values.

Additionally, I fit a log-normal distribution to the positive Loss values. The fitted curves and fitting results are shown in Figure A.5 and Table A.2. The fitted curves also appear to describe the observed frequencies well, but the RMSE is larger, while the MAE is similar.

Overall, the geometric distribution provides a better fit to the observed positive Loss frequencies in terms of RMSE, with an estimated parameter $p \approx 0.17$.

4.2.4 Length

The observed Length frequencies do not show an approximately linear pattern on either the semi-log or log-log plot. I therefore explore a log-normal distribution as a candidate model. Using the same fitting process as for Area, I obtain the fitted curves shown alongside the observed positive Length frequencies in Figure 4.6.

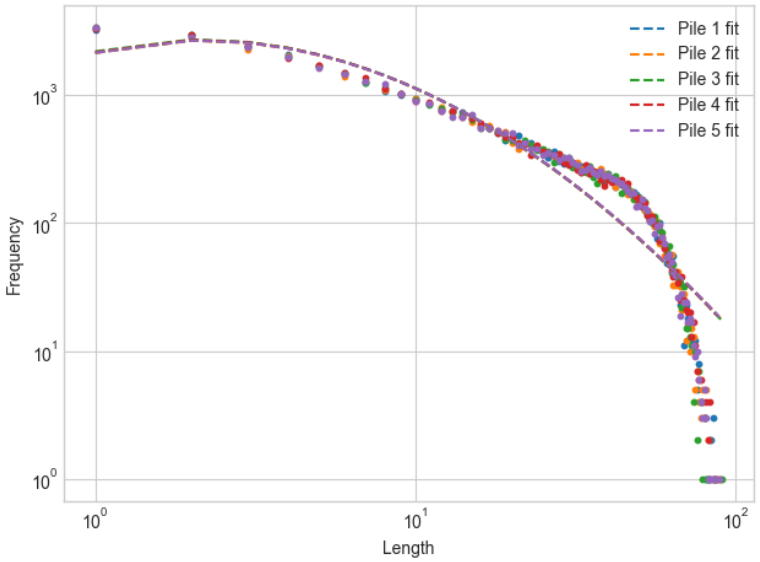


Figure 4.6: Observed positive Length frequencies and fitted log-normal curves for five independent simulations on log-log axes.

According to the figure above, there is a noticeable discrepancy between the fitted frequencies and the observed frequencies, though the fitted frequencies do provide a rough indication of the trend in the observed data.

Similarly, I calculate and record the estimated parameters μ and σ^2, together with MAE and RMSE for each pile, as shown in Table 4.4.

Pile	μ	σ^2	MAE	RMSE
1	2.1438	1.4079	89.6531	167.0913
2	2.1475	1.4166	93.1109	176.6598
3	2.1334	1.4183	90.0540	169.1877
4	2.1410	1.4138	86.4604	167.3350
5	2.1483	1.4174	88.8975	170.9569

Table 4.4: Estimated log-normal parameters and fitting errors for positive Length values.

Overall, the log-normal model with $\mu \approx 2.14$ and $\sigma^2 \approx 1.41$ provides a rough approximation to the distribution of positive Length values, although noticeable discrepancies remain.

4.3 Correlations Between Avalanche Properties

In order to explore the relationships between the four avalanche properties, I first generate a pairwise scatter plot for Pile 1.

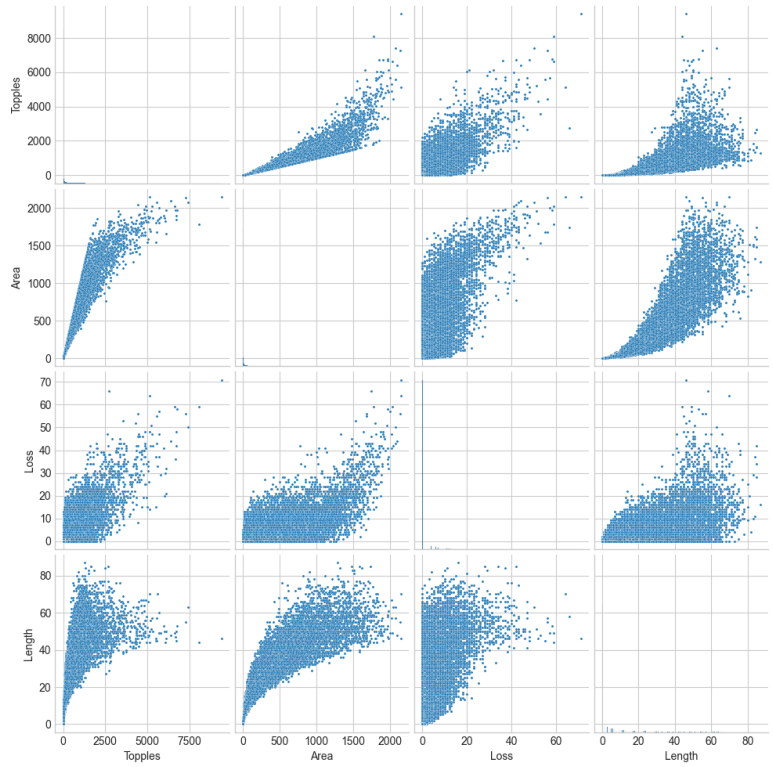


Figure 4.7: Pairwise scatter plot of the four avalanche properties for Pile 1.

As shown in Figure 4.7, the relationships between the properties appear to be positive, with some showing non-linear patterns.

Therefore, I calculate the Spearman correlation coefficients for Pile 1 to quantify the strength of the monotonic relationships between the properties.

	Topples	Area	Loss	Length
Topples	1.000000	0.999919	0.626868	0.961980
Area	0.999919	1.000000	0.626550	0.962282
Loss	0.626868	0.626550	1.000000	0.626531
Length	0.961980	0.962282	0.626531	1.000000

Table 4.5: Spearman correlation matrix for Pile 1.

The correlation matrices for Piles 2 to 5 are shown in Tables A.3, A.4, A.5, and A.6 in Appendix A. The correlation coefficients for the same pair of properties are similar across all five piles.

Overall, Topples has very strong positive correlations with Area and Length, with coefficients above 0.9. Area and Length are also strongly positively correlated. Loss has lower positive correlations with the other properties, with coefficients around 0.63.

Chapter 5

Extended model

5.1 Model Rules

In the extended model, toppling depends on the height difference between neighbouring sites rather than the height of an individual site. I set $k = 4$, so a site transfers one grain of sand to each nearest neighbour whose height is at least four grains lower. All four directions are checked using the site's height before any grains are transferred.

Additionally, the height outside the grid is fixed at zero. Therefore, a boundary site with a height of at least four transfers one grain out of the system in each outward direction. Sites are processed row by row, from left to right within each row, and the process is repeated until the configuration is stable.

For example:

$$\begin{pmatrix} 0 & 4 & 1 \\ 1 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 \\ 1 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

5.2 Simulation and Results

Similarly, I initialise a 50×50 matrix with all elements set to zero and initially set T=100,000. For the extended model, I simulate only one pile. I also set the random seed to 1643210 to ensure reproducibility. After each grain addition and the completion of all toppling activity, I record the average height, Topples, Area, Loss and Length, as in the basic model, for comparison.

I first plot the average height against the number of grains added to estimate when the system reaches a steady state. After 100,000 grains have been added, the average height is still increasing, as shown in Figure A.6. As the number of grains added to the system increases, the average height continues to rise, but at a decreasing rate. I generate a heat map to examine the spatial distribution of sand.

As shown in Figure A.7, boundary sites contain fewer grains because their heights must remain below four after all toppling activity has ceased. Heights generally increase towards the centre, forming a flat-topped pyramid. I expect that further grain additions will raise the central region and reduce the size of the flat top, producing a shape closer to a pyramid. To investigate this, I add a further 100,000 grains of sand to the grid.

As shown in Figure 5.1, the average height levels off at approximately 26.5 after about 190,000 grain additions. Therefore, I exclude the first 190,000 records as burn-in.



Figure 5.1: Average pile height against the number of grains added (left), with a zoomed-in view of the final stage (right).

After a further 100,000 grains are added, the heat map shows a shape closer to a pyramid, consistent with the earlier prediction, as shown in Figure 5.2.

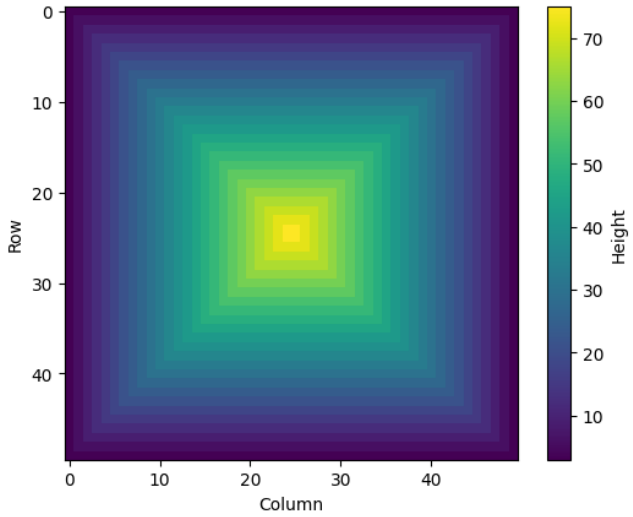


Figure 5.2: Spatial distribution of sand heights in the extended model.

After excluding the records collected during the burn-in period, I use the remaining records to plot the observed frequencies of the four avalanche properties and explore their distributions.

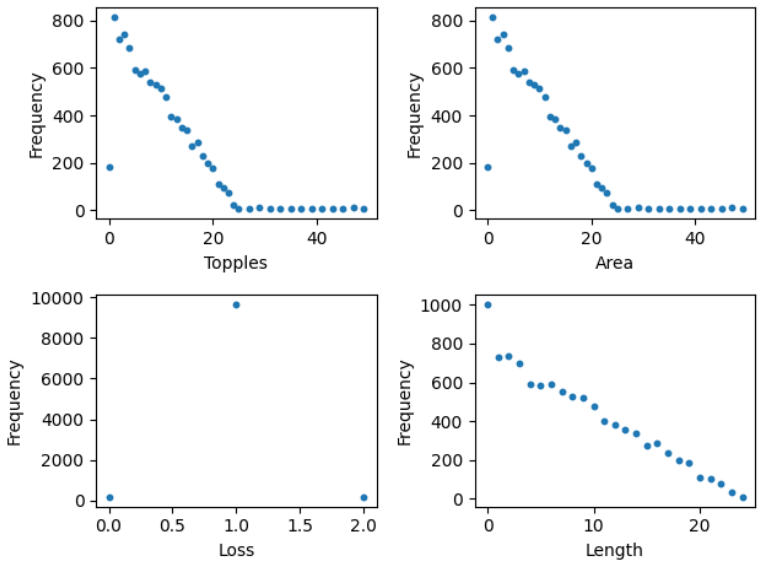


Figure 5.3: Observed frequencies of Topples, Area, Loss, and Length for one simulation of the extended model on linear axes.

As shown in Figure 5.3, the frequencies of Topples and Area decrease approximately linearly over the main range of positive values, with a small number of larger events forming a tail. Length also shows an approximately linear decreasing trend. Additionally, 9,622 of the 10,000 Loss values recorded after the burn-in period are equal to 1, while 185 are equal to 0 and 193 are equal to 2.

Chapter 6

Discussion

6.1 Steady State

Since each site contains at most three grains after all toppling activity has ceased, the average pile height cannot exceed three. Starting from an empty grid, I expect the average height to increase initially. As sand accumulates, toppling at the boundaries allows grains to leave the system, reducing the average height, while further grain additions increase it. Eventually, I expect the average height to fluctuate around a stable value, as the average loss of sand balances the input.

According to Figure 4.1, the average height initially increases as the number of grains added increases. By about 10,000 grain additions, the average height fluctuates around 2.1, consistent with the expected steady-state behaviour.

A process is at steady state if all properties of the system remain constant over time [4]. A process being in a statistically stationary state usually means that the statistical properties of the system, such as its distribution, do not change over time. In an equilibrium state, the system has no potential energy driving force for change, and all opposing forces are counterbalanced, so the properties of the system do not change over time [4].

I expect the BTW model to reach a statistically stationary state. According to the definition of equilibrium above, a system in equilibrium has no driving force for change. However, the BTW model is continuously driven by sand addition, with sand lost through the open boundaries. I expect the average height to fluctuate around a stable value, which is consistent with statistically stationary behaviour. Therefore, the average height can be used to characterise this state.

6.2 Sampling and Ergodicity

In general, sampling a system over time means recording the properties of interest from the same system at different times. These records can be used to study how the properties vary over time and to estimate their statistical distributions. On the other hand, sampling an ensemble of systems means recording the properties of interest from many systems of the same model at a fixed time[6].

A dynamical system is ergodic if its long-term time averages equal its space averages for all initial conditions. In particular, sampling an ergodic system over a sufficiently long time can yield the same statistical results as sampling an ensemble of systems [6].I expect the BTW model to be ergodic in its statistically stationary state because random sand addition and repeated toppling continually change the arrangement of sand, allowing a single pile to sample many different arrangements over time.

6.3 Correlation

For the four avalanche properties recorded, I expect positive correlations between them. In particular, Topples, Area, and Length may have stronger correlations with one another than with Loss. Based on their definitions, avalanches involving more sites and spreading farther generally require more toppling events. Loss may have weaker correlations with the other properties because it only increases when toppling occurs at the boundary.

The Spearman correlation coefficients obtained from the simulations support this opinion. As shown in Table 4.5, the correlations between Topples and Area, Topples and Length, and Area and Length are all greater than 0.9, while the correlations of Loss with the other three properties are approximately 0.63.

6.4 Extended Model

One noticeable difference is the spatial distribution of sand. The extended model forms a pyramid-shaped pile, with heights generally increasing from the boundaries towards the centre. The basic BTW model exhibits a disordered spatial distribution. Although sand is added at randomly selected sites in both models, the extended model develops a clear spatial pattern.

This difference is due to the change in the toppling rules. In the extended model, sand leaves the system when a boundary site contains four or more grains. After all toppling activity has ceased, each boundary site therefore contains at most three grains. A site immediately next to the boundary can contain at most six grains without toppling. Similarly, sites farther from the boundary can support greater heights without toppling, allowing more sand to accumulate towards the centre.

As more grains are added, the heights in the central region increase, while boundary heights remain at most three after stabilisation. This helps explain the pyramid shape observed in the simulation. The greater amount of sand that the extended model can retain also helps explain why more grain additions are needed before the average height levels off, which occurs after approximately 190,000 additions in this simulation.

After 190,000 grain additions, 9,622 of the remaining 10,000 observations record Loss = 1, meaning that the average height remains unchanged after a grain is added and all toppling activity has ceased. Loss = 0 and Loss = 2 are observed 185 and 193 times, respectively. Therefore, over these final 10,000 additions, the total number of grains in the system decreases by only eight, corresponding to a decrease of 0.0032 in the average height.

For the basic BTW model, the observed Topples frequencies show approximate power-law behaviour, appearing approximately linear on the log-log plot. For the extended model, the Topples frequencies decrease approximately linearly on linear axes up to about 25, followed by low frequencies for larger values. Area shows a similar pattern, and the frequency distributions of Topples and Area are nearly the same. Since Topples counts all toppling events while Area counts distinct toppled sites, this similarity suggests that repeated toppling at the same site within an avalanche may be uncommon.

It is also worth noting that, unlike in the basic model, the frequency at zero for both Topples and Area is much lower than the frequencies of some small positive values. This indicates that certain small avalanches occur more frequently than events with no toppling under the extended toppling rules.

Length also shows an approximately linear decrease in frequency over the observed range, differing from the pattern observed in the basic model.

6.5 Limitations and Further Studies

First, this project only simulates grids of one size, which is 50×50. How the grid size affects the distributions of system properties is not investigated. Grids of different sizes could be tested to explore whether and how grid size affects these properties.

Second, the distribution-fitting process only uses plots, MAE and RMSE to evaluate the fits. Some discrepancies remain between the fitted and observed frequencies, particularly for Length, for which the fitted distribution provides only a rough approximation. Further statistical tests could be used to examine the fits, and additional candidate distributions could be explored.

Finally, only one pile is simulated for the extended model. A large number of grain additions is needed for the average height to stabilise, making the simulation time-consuming. More independent simulations could help determine whether the observed properties are consistent across different runs.

Chapter 7

Conclusion

This report focused on simulating the basic BTW model, exploring its steady-state behaviour and the frequency distributions of four avalanche properties, and examining its Abelian property. An extended model with gradient-dependent toppling rules was also simulated, and its steady-state behaviour and avalanche frequency distributions were compared with those of the basic model.

For the basic BTW model, the average height fluctuates around 2.1 after about 10,000 grain additions. The distribution of Topples exhibits approximate power-law behaviour. The distribution of positive Area values is reasonably approximated by a log-normal distribution, while positive Loss values are approximately described by a geometric distribution. A log-normal distribution provides only a rough approximation for positive Length values. Spearman correlations were also calculated between the four properties. Topples, Area and Length are strongly positively correlated with one another, with coefficients above 0.9. In contrast, Loss has weaker positive correlations with these properties, with coefficients of approximately 0.63.

For the extended model, the average height fluctuates around 26.5 after about 190,000 grain additions. The spatial distribution of sand forms a pyramid-shaped pile, with heights generally increasing towards the centre. The frequencies of Topples and Area decrease approximately linearly up to about 25, followed by a tail of low-frequency events. Length also shows an approximately linear decrease in frequency, without a similar tail in the observed range. The observed Loss values are limited to 0, 1 and 2, with Loss = 1 accounting for 96.22% of the 10,000 observations after burn-in.

Overall, the comparison shows that changing the local toppling rules can largely affect both the spatial distribution of sand and avalanche statistics, even when the average height in both models eventually stabilises.

Appendix A

Supplementary Results

A.1 Linear-scale Frequency Distributions

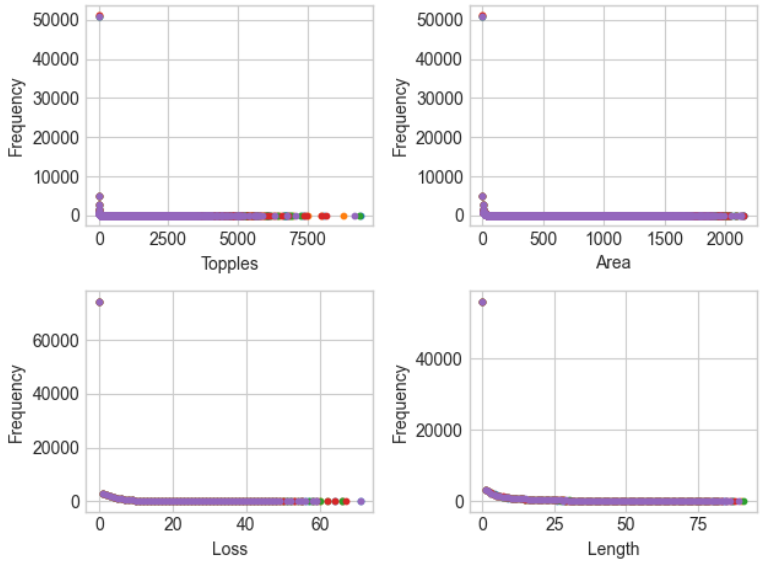


Figure A.1: Observed frequencies of Topples, Area, Loss, and Length for five independent simulations on linear axes.

A.2 Spatial Distribution of Sand Heights

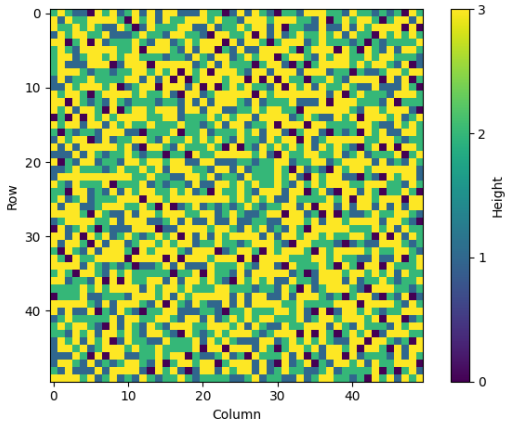


Figure A.2: Spatial distribution of sand heights in the final configuration of Pile 5 of the basic model.

A.3 Additional Distribution Fits

A.3.1 Power-law Fit For Area

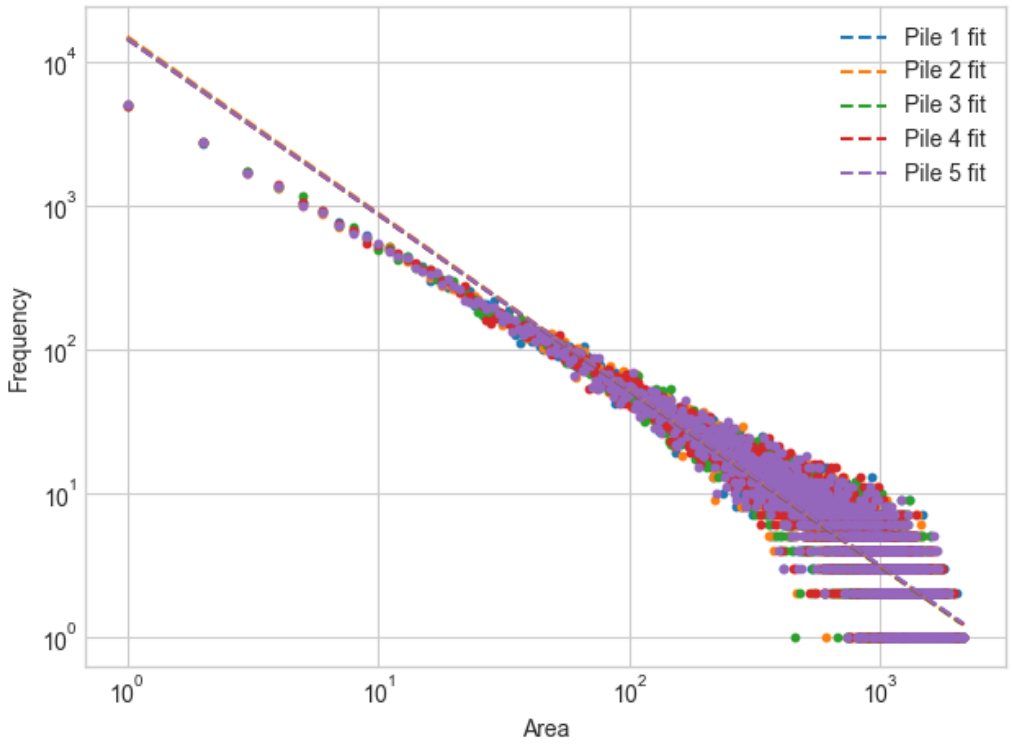


Figure A.3: Observed Area frequencies and fitted power-law curves for five independent simulations on log-log axes.

Pile	α	MAE	RMSE
1	-1.2198	15.7039	252.2666
2	-1.2276	16.9752	272.2115
3	-1.2251	15.8027	259.4345
4	-1.2215	15.6431	254.8015
5	-1.2206	16.1838	257.3404

Table A.1: Power law exponents and fitting errors for the power-law fit to Area.

A.3.2 Geometric Fit for Loss Including Zero

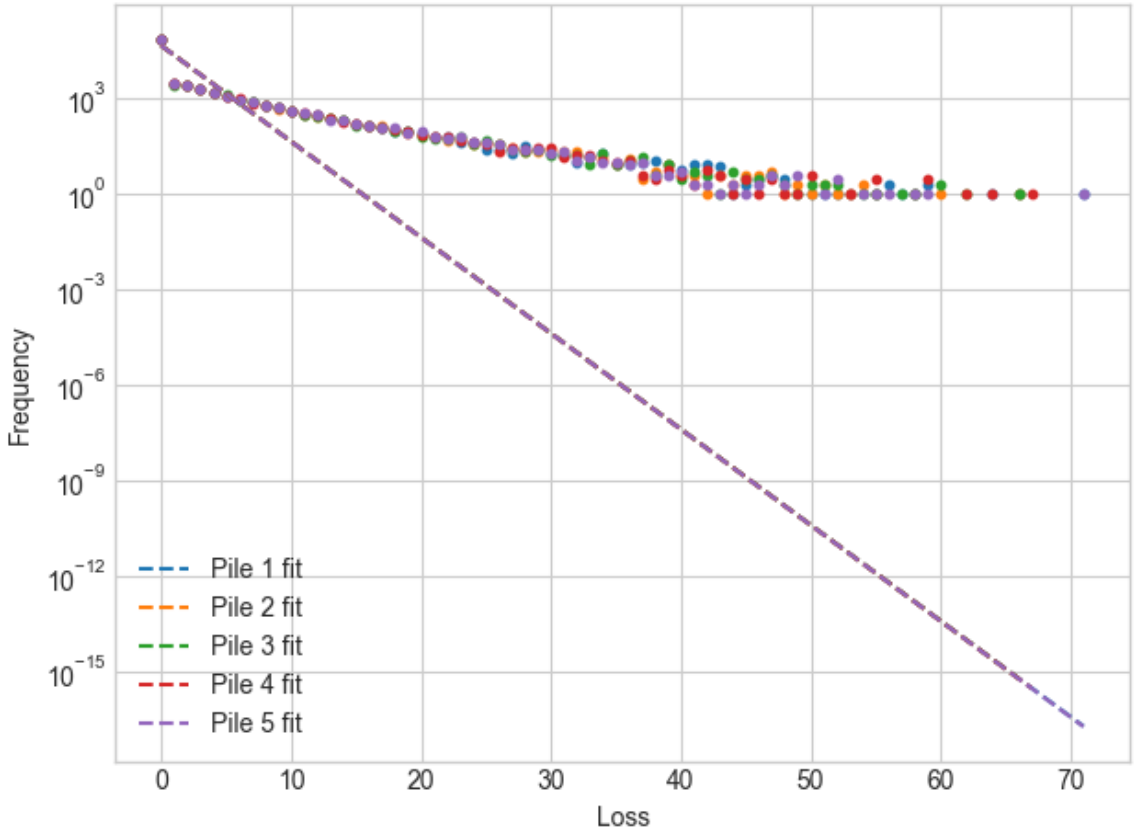


Figure A.4: Observed Loss frequencies and fitted geometric curves including zero for five independent simulations on semi-log axes.

A.3.3 Log-normal Fit for Loss

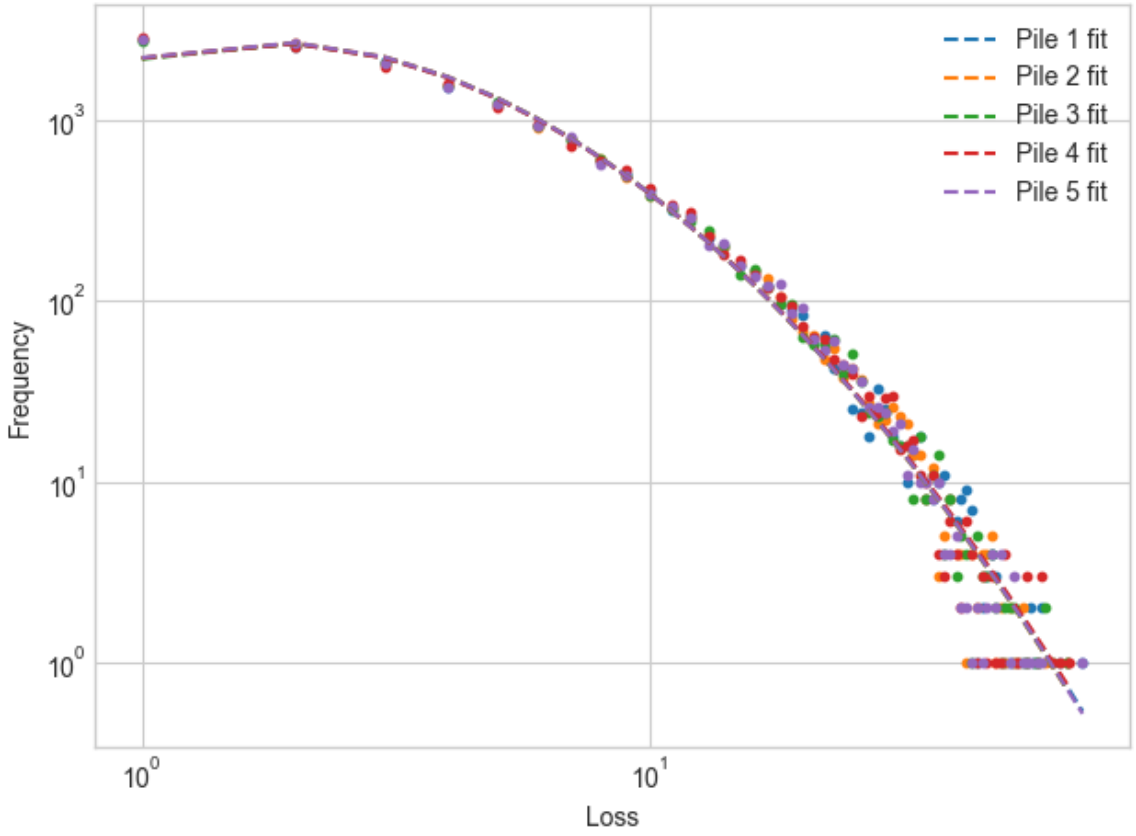


Figure A.5: Observed positive Loss frequencies and fitted log-normal curves for five independent simulations on log-log axes.

Pile	μ	σ^2	MAE	RMSE
1	1.3347	0.8261	26.2768	88.9550
2	1.3333	0.8187	24.9234	83.5375
3	1.3377	0.8179	24.4259	83.0422
4	1.3361	0.8304	30.6826	96.2659
5	1.3321	0.8215	27.6485	88.4346

Table A.2: Estimated log-normal parameters and fitting errors for positive Loss values.

A.4 Additional Correlation Matrices

A.4.1 Correlation Matrix for Pile 2

	Topples	Area	Loss	Length
Topples	1.000000	0.999918	0.631324	0.962060
Area	0.999918	1.000000	0.630935	0.962363
Loss	0.631324	0.630935	1.000000	0.630655
Length	0.962060	0.962363	0.630655	1.000000

Table A.3: Spearman correlation matrix for Pile 2.

A.4.2 Correlation Matrix for Pile 3

	Topples	Area	Loss	Length
Topples	1.000000	0.999921	0.627074	0.962685
Area	0.999921	1.000000	0.626732	0.962977
Loss	0.627074	0.626732	1.000000	0.625866
Length	0.962685	0.962977	0.625866	1.000000

Table A.4: Spearman correlation matrix for Pile 3.

A.4.3 Correlation Matrix for Pile 4

	Topples	Area	Loss	Length
Topples	1.000000	0.999919	0.627299	0.962661
Area	0.999919	1.000000	0.627074	0.962962
Loss	0.627299	0.627074	1.000000	0.626932
Length	0.962661	0.962962	0.626932	1.000000

Table A.5: Spearman correlation matrix for Pile 4.

A.4.4 Correlation Matrix for Pile 5

	Topples	Area	Loss	Length
Topples	1.000000	0.999916	0.629966	0.961779
Area	0.999916	1.000000	0.629602	0.962091
Loss	0.629966	0.629602	1.000000	0.630056
Length	0.961779	0.962091	0.630056	1.000000

Table A.6: Spearman correlation matrix for Pile 5.

A.5 Extended Model

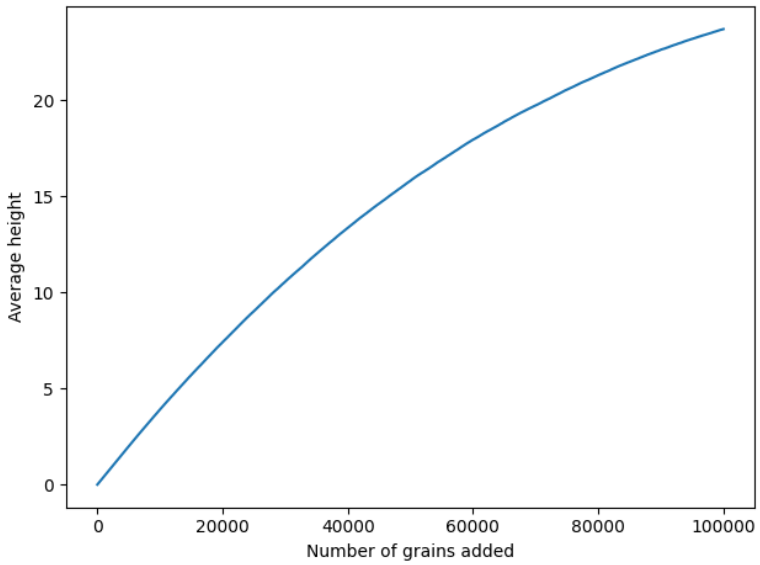


Figure A.6: Average pile height against the number of grains added.

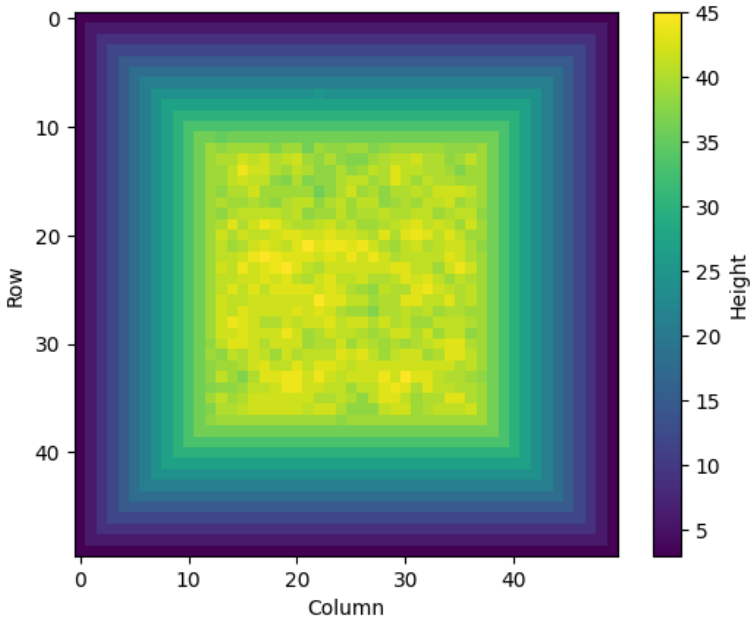


Figure A.7: Spatial distribution of sand heights in the extended model.

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